



# Investigation report

## Rejected take-off and runway overrun, Tarlton Springs - Standard investigation

Drafted 07/10/2026

INVESTIGATION WORKFLOW SUITE

## Contents

|                                                    |    |
|----------------------------------------------------|----|
| Executive summary .....                            | 3  |
| What happened.....                                 | 3  |
| Outcome .....                                      | 3  |
| Why it happened .....                              | 3  |
| Safety issues .....                                | 4  |
| Safety action.....                                 | 4  |
| Event overview .....                               | 5  |
| The flight crew and the duty.....                  | 5  |
| Conditions and preparation for departure.....      | 5  |
| The take-off roll and the rejected take-off.....   | 5  |
| The strike and the stop .....                      | 6  |
| Immediately after the stop .....                   | 6  |
| Analysis .....                                     | 8  |
| Introduction .....                                 | 8  |
| People .....                                       | 8  |
| Environment .....                                  | 10 |
| Equipment .....                                    | 13 |
| Procedures .....                                   | 14 |
| Organisation.....                                  | 16 |
| Hypotheses tested and excluded .....               | 18 |
| Findings summary.....                              | 20 |
| Contributing factors .....                         | 20 |
| Other factors that increased risk .....            | 20 |
| Other findings.....                                | 20 |
| Appendices.....                                    | 22 |
| Appendix A: Test summary for analysed factors..... | 22 |
| Appendix B: Excluded hypotheses.....               | 23 |
| Appendix C: Sources of evidence.....               | 24 |
| Appendix D: Investigation activity .....           | 25 |
| Distribution .....                                 | 25 |

## Executive summary

### What happened

On 18/06/2026 at about 17:30 local, Saltbush Air SB512, VH-SBQ, a 50-seat twin-turboprop, rejected its take-off from runway 09 at Tarlton Springs Aerodrome. The roll began in darkness, five minutes after last light, with no moon. The first officer, who was flying, saw two kangaroos crossing onto the sealed runway. The captain rejected the take-off at 118 kt, 6 kt above V1. One second later the aircraft struck both animals, and the No. 2 reverse was lost. The aircraft left the runway end at about 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area.

### Outcome

On board were 44 passengers, 2 flight crew and 1 cabin crew member. No evacuation was commanded, and passengers left by the forward airstair. There were no serious injuries. One cabin crew member reported bruising sustained during disembarkation.

The aircraft was substantially damaged:

- three No. 2 propeller blades were beyond limits, and the engine needed a shock-load inspection
- the right main gear doors were distorted
- both right main tyres deflated through the fuse plugs at 84% of certified brake energy
- the lower right fuselage was punctured

Two kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes. The aircraft stopped 28 m short of the end of the prepared safety area, beyond which the terrain is ungraded. The operator suspended the evening Tarlton Springs service.

### Why it happened

Two causes acted on the same stop, and neither alone took the aircraft past the runway end. Because the reject began after V1, 695 m of runway remained against the 757 m needed. The strike then left reverse on No. 1 only, so deceleration was about 15% below the certified value.

The captain judged that rotating into two large animals at night was the greater risk, and acknowledged the decision was outside procedure. Neither training nor the operations manual had ever presented a crew with a runway obstruction. It is likely that this gap contributed to the decision. The investigation also weighed the captain's account that he knew the rule and departed from it deliberately.

The animals entered through a failed section of the northern perimeter fence, crossed by a clear animal pad 138 m before the first carcass. The aerodrome operator had recorded the failure in 04/2023, with animals seen passing through, and deferred its repair unfunded at three budget rounds. The departure fell within the hours when the aerodrome's sighting log records almost all kangaroo activity, and in darkness the strip could not be seen from the centreline.

No runway inspection was requested, although the aerodrome would have carried one out. No document prompted either crew member to consider one. The operator required an inspection after last light only where a company wildlife strike had been reported, a condition met only by the event the inspection exists to prevent.

The evening service was approved as a schedule change only, so the route risk assessment restricting Tarlton Springs to daylight operations was not reopened. Below the aerodrome's size,

Aerodrome Standard AS-139 did not require a documented wildlife hazard management plan, and none was held. The aerodrome was not in breach. The investigation found it likely that the absence of a plan contributed, but held this finding less firmly than the other organisational findings.

The captain established maximum braking 2 seconds after the decision, against a fleet mean of 3.2 seconds, which reduced how far the aircraft travelled into the safety area.

## Safety issues

Five safety issues contributed:

- the regulatory threshold for a wildlife plan
- the unrepaired fence
- the inspection trigger
- the absence of obstruction training and guidance
- the approval of the evening service without risk assessment review

Two further safety issues did not contribute but increased risk. The evening rotation is rostered to close five-sector duties within 45 minutes of the limit. Floor escape path lighting is checked only every 24 months, an interval that had hidden two dark segments on VH-SBQ.

Two matters could not be resolved: whether the crew were fatigued, and whether four edge lights measured above nominal intensity affected when the animals were seen.

## Safety action

On 26/06/2026 the operator listed Tarlton Springs in Appendix 8A by temporary revision TR-2026-04, requiring an inspection before every departure there after last light. The investigation has recommended that the operator list aerodromes on wildlife exposure rather than a reported strike, and adopt a route risk assessment trigger for any aerodrome first operated into after last light. A response is due by 30/09/2026. The aerodrome operator installed a temporary electric wire on 25/06/2026 and has funded fence replacement for 09/2026.

## Event overview

On 18/06/2026, Saltbush Air SB512, VH-SBQ, a 50-seat twin-turboprop, was scheduled to depart Tarlton Springs Aerodrome for Port Carrick on the last scheduled service of the day. Times are UTC unless stated otherwise. Local time was 10 hours ahead, so the occurrence took place at about 17:30 local.

The only runway inspection that day was carried out at 21:35 on 17/06/2026 (07:35 local on the day of the occurrence), before the first scheduled arrival. It recorded the movement area as clear. No further inspection was carried out or requested.

## The flight crew and the duty

The pilot in command had 8,410 hours total, 2,140 on type, and 3 years 1 month in command. The first officer, pilot flying for SB512, had 1,655 hours total and 405 on type. She had flown two previous sectors into Tarlton Springs, both in daylight, on 09/01/2026 and 17/02/2026.

The crew signed on at 06:30 local for three sectors: Port Carrick to Hensby, Hensby to Mardle Creek, and Mardle Creek to Port Carrick. That morning a member of the rostered crew for the evening Tarlton Springs rotation reported unfit, and the rotation was added to this crew's duty. SB512 became the fifth sector. The occurrence came 11 hours 00 minutes into a duty planned at 13 hours 00 minutes, against a maximum of 13 hours 30 minutes under the company's approved fatigue risk management system. A ground delay at Port Carrick was recovered on the Tarlton Springs turnaround. The doors closed at 17:16 local and the flight departed on schedule with 44 passengers, 2 flight crew and 1 cabin crew member.

## Conditions and preparation for departure

The take-off roll began in darkness, five minutes after last light, with no moon. Tarlton Springs has no control tower, so the crew made their own CTAF broadcasts.

At 07:21:50 the pilot in command activated the pilot-activated runway lighting with three transmitter keyings, remarking "Lights are on. That's thirty minutes on the timer." The crew broadcast taxiing for runway 09 at 07:23:15. At 07:28:51 the first officer read the AWIS wind as 090 degrees at 6 kt, and the lining-up broadcast followed at 07:29:10. No runway inspection was requested or offered, and no wildlife transmission appears on the CTAF in the preceding 90 minutes. The pilot in command then said "Right, call's made, lights are on, nobody else about. Your aeroplane, your take-off."

V1 was calculated as 112 kt, based on 20,800 kg, flap 10, a dry runway 09 and a 5 kt headwind component.

## The take-off roll and the rejected take-off

At 07:30:02 the first officer called "Power set", with torque at 98.4% and 98.5%. The pilot in command confirmed "Power set, both engines good." The 80 kt cross-check at 07:30:15 showed no caution or warning. The pilot in command called V1 at 07:30:29, at 112 kt indicated. The recorded headwind component was 4 to 6 kt throughout the roll.

At 07:30:31, at 116 kt, the first officer called "Roos - roos, roos, on the left, they're on the -". The occurrence notification records two kangaroos moving onto the runway from the northern side, past the 1,100 m point. The first officer estimated their range at first sight at around 200 m. After the stop she said "They came out of the dark on the left of the strip, crossed the centreline," and

that she had them in view for "two seconds. Maybe less." The pilot in command said on the CVR that he "never had them" and had been "heads-in on the engine page after V one."

At 07:30:32 the pilot in command called "Stopping. I have control," and the first officer replied "You have control." Both power levers reached flight idle within 0.6 seconds. The reject began at 118 kt, 6 kt and 3 seconds past V1, with the aircraft 1,152 m along runway 09 and 695 m of sealed surface remaining.

In his air safety report, the pilot in command stated his judgement that continuing would put the aircraft airborne at night with damage that could not be assessed, over country offering no usable option. He acknowledged that the decision was taken above V1.

## The strike and the stop

At 07:30:33 the recorders captured a sound consistent with a heavy impact, lasting about 0.4 seconds and followed by airframe rumble. The flight data showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The sources differ on the number of animals struck:

- the occurrence notification and the post-occurrence runway survey record that both animals were struck, and the survey recovered carcasses at 1,188 m and 1,221 m from the runway 09 threshold
- the pilot in command's account records striking at least one animal

The right main landing gear and the No. 2 propeller took the impact. Three No. 2 propeller blades were damaged beyond repairable limits, and animal tissue was recovered from two blade roots.

Maximum braking was established at 07:30:34, 2 seconds after the decision, with anti-skid active on all four wheels. The operator's fleet mean for that interval in rejected take-offs is 3.2 seconds. At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse was obtained on No. 1 only, because No. 2 propeller RPM had been decaying since the impact and the No. 2 reverse interlock did not make.

Mean deceleration was 2.6 m/s<sup>2</sup> from 07:30:34 to 07:30:44, while the first officer called the speed down, "Sixty ... fifty ...". At 07:30:47 the pilot in command said "We're not stopping on it." Two seconds later the first officer called "End's coming up."

The aircraft left the end of runway 09 at 07:30:51 at about 21 kt, and deceleration fell to 1.9 m/s<sup>2</sup> in the graded gravel. Continuous tyre marks ran from 1,160 m to the runway end. The aircraft came to rest at 07:30:57, upright on the extended centreline, 62 m into the 90 m runway end safety area and 28 m short of the end of the prepared area. The stopping distance from 118 kt was 757 m.

## Immediately after the stop

At 07:30:59 the pilot in command asked "OK. We're stopped. Everyone alright?" He set the parking brake, noted no fire indications and stated "I'm not evacuating." The first officer agreed. Both engines were shut down at 07:31:26, and at 07:31:34 he told the cabin to "remain seated, remain seated."

At 07:31:48 the aerodrome reporting officer, who had seen the aircraft go off the end, called on the CTAF. The pilot in command asked for the aerodrome emergency plan to be activated and a vehicle sent. The aerodrome reporting officer activated the plan at 07:32:10, dispatched a vehicle, and at 07:32:24 broadcast that runway 09/27 was obstructed and the aerodrome closed.

No evacuation was commanded and no slides were deployed. Passengers disembarked by the forward airstair, heard operating at 07:34:40. The sources record the timing differently. The occurrence notification states the disembarkation took place 4 minutes 8 seconds after the aircraft

stopped, while the timeline records it running from 07:34:40 to 07:38:48, a duration of 4 minutes 8 seconds. The main cabin lights were on throughout. The first officer later described sections of the aisle floor strip towards the rear as unlit while the emergency lighting was armed, and raised this with the engineer the next morning.

There was no fire, and no injuries were reported at the time. One cabin crew member later reported bruising sustained during the disembarkation. Two kangaroos were killed. Damage to the aircraft included:

- three No. 2 propeller blades beyond limits, and a shock-load inspection on that engine
- distorted right main gear doors
- both right main tyres deflated through the fuse plugs, with brake energy at 84% of the certified maximum
- abrasion and two punctures to the lower right fuselage skin

The aerodrome closed for 4 hours 20 minutes. The operator's engineering representative removed the FDR and CVR at 12:40 UTC. Both flight crew were stood down and drug and alcohol tested, with negative results. The Head of Flying Operations suspended the evening Tarlton Springs service.

# Analysis

## Introduction

The overrun had two physical causes that acted on the same stop. The take-off was rejected 3 seconds and 6 kt after V1, which shortened the runway left for the stop. One second later the aircraft struck the two kangaroos, which removed the No. 2 reverse and lowered the deceleration achieved. Neither alone took the aircraft past the end of runway 09. This analysis works back from those two events to the conditions and organisational arrangements that put two kangaroos on the sealed surface in darkness, with no inspection beforehand, in front of a crew who had never been prepared for that situation. It also examines two matters that could not be settled, two conditions that increased risk without contributing, and two hypotheses that were tested and excluded.

## People

### The decision to reject above V1

The FDR shows V1 called at 112 kt at 07:30:29. Both power levers reached flight idle within 0.6 seconds of 07:30:32, at 118 kt. Operations Manual Part B 4.6.4 is unqualified: "At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached." The CVR shows that the whole decision took one exchange. The first officer's sighting call came at 07:30:31, and the captain's "Stopping. I have control." followed at 07:30:32.

The captain gave the same account in interview and in his air safety report. He judged that rotating into two large animals would leave the aircraft airborne at night with damage that could not be assessed, over country that offered no usable option. He chose the risk whose outcome he could see. He also stated plainly that the decision was not in accordance with the procedure. The departure from 4.6.4 was therefore deliberate and made in full knowledge of the rule, not through any misunderstanding of it.

The consequence was fixed by the performance schedule. The scheduled accelerate-stop distance of 1,780 m against an ASDA of 1,847 m left a margin of 67 m. That margin assumes the stop begins at or before V1. Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along runway 09 at the moment of the reject, with 695 m of sealed surface remaining. A reject begun late does not reduce the deceleration available. It reduces the runway on which that deceleration has to work, and the 6 kt and 3 seconds past V1 took the stop outside the conditions the schedule was built on.

This decision was taken against a background examined later in this section and under Procedures. The animals were on the runway at all, the darkness limited when they could be seen, and neither the operator's training nor its procedures had ever presented an obstruction on the runway. Those conditions explain how the decision came to be made in about a second. They do not make it a different decision.

It is virtually certain that the take-off was rejected 6 kt above V1, and extremely likely that this contributed to the runway overrun.

### No runway inspection requested before departure

Neither the CVR nor the Tarlton Springs CTAF recording contains a request for a runway inspection, an offer of one, or any mention of wildlife before the departure. The captain stated that wildlife was not discussed with anyone during the turnaround. The aerodrome's daily inspection log for

18/06/2026 shows one inspection, at 07:35 local, before the first arrival, with the movement area recorded as clear. No later inspection was carried out or requested.

An inspection was available. Aerodrome Manual 6.3 provides for inspections outside the scheduled time at the request of the reporting officer or a pilot in command. The reporting officer was on duty at the terminal throughout the turnaround and confirmed that an inspection would have been carried out if asked. SB512 was the only departure after last light that day.

Both crew members stated that they did not consider requesting one. The captain had expected Tarlton Springs to be listed in Appendix 8A, and was surprised to learn that it was not. Nothing in the documents the crew worked to asked either of them to consider an inspection at this aerodrome, for the reasons set out under Procedures. The omission is therefore best understood as a step that no procedure or document put in front of the crew, rather than a step the crew overlooked.

The influence of the omission follows from what an inspection would have covered. An inspection would have included the section from 1,050 m to 1,250 m, where the animal pad through the fence and both carcasses were found, and aerodrome staff disperse animals when they find them. The case falls short of certainty because an inspection completed some minutes before the roll cannot guarantee that the runway stays clear until the aircraft passes.

It is virtually certain that no runway inspection was requested before SB512 departed after last light, and likely that this contributed to the two kangaroos being on the sealed runway during the take-off roll.

### **The response once the reject was initiated**

The FDR shows the power levers at flight idle 0.6 seconds after the decision and maximum braking established at 07:30:34, 2 seconds after it. The operator's recorded fleet mean from decision to maximum braking in rejected take-offs is 3.2 seconds. Anti-skid was active on all four wheels throughout. Brake energy reached 84% of the certified maximum without a brake fire. Reverse was selected on No. 1 at 07:30:36, 4 seconds after the decision. No. 2 was not available, because its propeller had been damaged in the strike. The first officer made the speed callouts and the call that the runway end was approaching.

On the reconstruction, a stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the 90 m RESA. Beyond that edge the terrain is ungraded. The aircraft instead left the seal at about 21 kt and stopped 62 m into a prepared surface, upright and on the extended centreline. The same investigation that finds the reject was outside procedure therefore also finds that the stop which followed was carried out faster than the fleet average, and that this materially limited the outcome.

It is virtually certain that maximum braking was established within 2 seconds of the reject decision, and very likely that this reduced the severity of the overrun.

### **Whether the flight crew were fatigued**

The investigation examined whether the flight crew were fatigued at the time of the take-off and could not settle it.

Several points support fatigue. The take-off came 11 hours into a duty that began with a 06:30 sign-on, on the fifth sector, at 17:30 local. That morning the day had been extended from three sectors to five when the crew took on the evening rotation from a crew member who reported unfit. In interview both crew members described themselves as tired at that point. The captain said that he would have declined another sector if asked.

Other evidence points the other way. Both crew members described the tiredness as ordinary and not impairing. The captain said he was not struggling and was fully engaged during the seconds that mattered. The first officer described a normal five-sector day, said she was not fighting tiredness, and said she had a proper break in Port Carrick in the middle of the duty. The duty was within the company's limits, planned at 13 hours against a 13 hour 30 minute maximum, after two rest days for the captain and one for the first officer. The recorded performance shows none of the usual signs of impairment. The response times described above are at the fast end of the fleet distribution, the callouts were made on time, and there is no error of omission anywhere on the roll.

The evidence for fatigue is the crew's own description of being tired, and being tired after 11 hours is not the same as being fatigued to a degree that degrades performance. The crew's fuller accounts and the recorded performance argue against impairment, and nothing else bears on the question. On that evidence, whether the crew were fatigued was assessed as about as likely as not. Settling it would require a detailed fatigue assessment, including each crew member's sleep history and duty pattern in the days before 18/06/2026. That assessment was not undertaken, because even if fatigue had been established it could not have been shown to influence the occurrence. Every mechanism through which fatigue would have had to act is contradicted by the recorded data. The one decision it could plausibly have affected, not requesting an inspection, was a decision no document prompted either crew member to make.

Whether the flight crew were fatigued at the time of the take-off remains unresolved, and it is not carried into the findings.

## Environment

### Two kangaroos on the sealed runway

Two adult kangaroos were recovered from runway 09, at 1,188 m and 1,221 m from the threshold, on the northern half of the runway. The first officer saw them crossing from the northern side towards the south, estimated their range at around 200 m at first sight, and called them at 07:30:31. The impact at 07:30:33 was with these two animals. The carcasses, the sighting call and the recorded impact each establish independently that the animals reached the sealed surface.

Their route onto the runway is a separate question, and it is examined under Organisation. The geometry is nonetheless relevant here. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass and on the same side of the runway (Figure 1). The aerodrome's sighting log holds eleven entries for kangaroos on the sealed surface of runway 09/27, all around last light. No inspection had been carried out since 07:35 local.



*Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026*

The first officer's call is the only trigger for the reject heard on the CVR. Both the reject and the strike follow directly from the animals being on the runway at that moment.

It is virtually certain that two kangaroos crossed the strip onto the sealed runway, and extremely likely that this contributed to the rejected take-off and to the strike.

### **Departure in the logged kangaroo peak**

The aerodrome's wildlife sighting log for 01/01/2025 to 18/06/2026 holds 46 entries. Of these, 41 record kangaroos on or adjacent to the movement area, and 38 of those 41 are timed between one hour before and two hours after last light. On 18/06/2026 last light was 17:25 local, so the window ran from 16:25 to 19:25. The roll began at 17:30. The captain's own experience matches the log. He has seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening and never in daytime.

On this evidence, two animals crossing the runway at 17:30 fits the pattern the aerodrome's own log describes, and was not a chance encounter. The scheduled departure fell inside the window whatever the timetable did that day.

Two qualifications apply. The log is a paper register whose completeness cannot be verified. It also records sightings rather than presence, and aerodrome staff go off duty at 18:30, part way through the window, so the concentration of entries may partly reflect when someone was there to see. Neither qualification undoes the pattern, but each limits the confidence that can be placed in it.

It is extremely likely that SB512 departed during the period of peak logged kangaroo activity at Tarlton Springs, and likely that this contributed to the two kangaroos being on the sealed runway during the take-off roll.

### **Darkness during the roll**

The roll began at 17:30 local (07:30:02 on the FDR), on schedule, 32 minutes after sunset and 5 minutes after the end of civil twilight. Moonrise was not until 22:41, and the moon was a 14% illuminated waning crescent. There was no natural ambient light on the strip and no town lighting within sight. The only light came from the runway edge lighting, which the crew had switched on through the pilot-activated lighting at 07:21:50. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip either side of the sealed surface cannot be seen from the centreline in those conditions.

In this light the first thing a crew can see of an animal crossing the strip is the animal arriving on the lit surface. The first officer's description on the CVR after the stop was that the animals "came out of the dark on the left of the strip". In interview she said that outside the lights it was black, and that the animals did not come gradually into view but were suddenly there. In daylight, two animals crossing the graded strip from the northern fence would have been visible before they reached the seal, and earlier in the roll.

The evidence on the other side is real. The first officer did see the animals, at around 200 m, once they were on the seal. How long they had been on the strip before that is not known, and animals that came onto the strip only seconds before reaching the seal would have been seen late in daylight too. The captain was heads-in on the engine page after V1, where the procedure places him, so in daylight the sighting would still have rested with the pilot flying, whose attention belongs on the centreline. These points hold the finding below the confidence of the other influence findings, because the animals' time on the strip cannot be established. They do not displace it. The site visit shows that nothing beside the seal could be seen in this light, and the first officer's account is of animals appearing out of the dark, not of animals she was slow to see.

The roll took place in darkness because of the schedule, not because of any delay. That link back to the change management decision is analysed under Organisation.

It is virtually certain that the take-off roll took place in darkness, 5 minutes after last light and with no moon, and likely that this contributed to the decision to reject the take-off above V1.

### **Edge lighting intensity in the section the animals crossed**

The investigation examined whether the intensity of four northern runway edge lights affected when the animals were seen. The condition itself was largely settled, but its effect could not be.

A photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of the nominal intensity for stage 3. These lights are in the section the animals crossed. The remaining lights were within tolerance. The runway lighting is pilot-activated. Once keyed from the CTAF it runs for 30 minutes at the single intensity set at the aerodrome's lighting controller, which was found at stage 3, the setting recorded at commissioning. Crews cannot vary it. The works register and electrical log show no work on the lighting or its controller between 18/06/2026 and the check. The aerodrome's maintenance programme checks lamp serviceability but does not measure intensity.

The measurement was taken seven days after the occurrence, and lamp output can change. However, the setting is fixed, nothing was done to the lighting in the interval, and nothing suggests the four lights were different on the night. On that basis it is likely that the four lights were at the measured intensity during the take-off, although nobody measured them at the time.

The case that the intensity mattered rests mainly on the captain's account. He recalled the lighting as brighter than he would have wanted, a corridor of light with everything outside it black, so that anything off the seal could not be seen coming. He also noted that with pilot-activated lighting there was nobody to ask to turn it down. The argument is that brighter edge lights deepen the contrast between the lit corridor and the dark strip beside it, which is where the animals came from.

The case against is stronger on the direct evidence. The first officer, the only crew member who saw the animals, described the lighting as normal. She said she saw them completely clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight landings at Tarlton Springs to compare against, she was not well placed to judge whether the lighting was unusual. The captain did not see the animals because he was heads-in on the engine page after V1, as the procedure requires, and that explanation does not involve the lighting at all. The strip either side of the seal cannot be seen from the centreline in darkness whatever the edge lights are doing.

The intensity is the settled part. The proposition that it delayed detection, and so affected the reject, is the unsettled part, and the available evidence does not reach it. The two crew accounts of the lighting disagree, the only witness to the animals saw them clearly on the seal, and darkness alone explains why nothing off the seal was seen. Resolving the question would require evidence linking an edge light at up to 131% of nominal intensity to the detection of an animal on or beside a runway, and the investigation had no such evidence. On the evidence available, whether the intensity influenced the occurrence was assessed as about as likely as not.

Whether the elevated intensity of the four northern edge lights affected the detection of the animals remains unresolved, and it is not carried into the findings.

## Equipment

### The strike and the loss of No. 2 reverse

The impact came at 07:30:33, at 114 kt, one second after the reject began. The FDR shows a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits, with leading edge deformation and one blade tip separated, and animal tissue was recovered from two blade roots. The right main gear doors were distorted, and the lower right fuselage skin was abraded and punctured by propeller debris.

The significance for the stop lies in the reverse. No. 2 propeller RPM decays from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, only No. 1 responded, as the captain's call on the CVR reflects. The strike therefore took away half of the reverse thrust at the start of a stop that already had no margin left.

It is virtually certain that the No. 2 propeller and right main gear struck the two kangaroos, and virtually certain that this contributed to the deceleration falling below the scheduled accelerate-stop performance.

### Deceleration below scheduled performance

The scheduled accelerate-stop figures assume symmetric reverse. With reverse on No. 1 only, the FDR shows a mean deceleration of 2.6 m/s<sup>2</sup> from 07:30:34 to 07:30:44. That is approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition. When the aircraft left the seal at 07:30:51, deceleration fell to 1.9 m/s<sup>2</sup> in the graded gravel. The stopping distance achieved from 118 kt was 757 m, against 695 m of sealed surface remaining. The FDR data, the distance reconstruction and the physical marks agree, with continuous tyre marks from 1,160 m to the runway end and the nose wheel rut ending 62 m into the RESA on the extended centreline.

At the certified deceleration, the reconstruction puts the stop from the same point at roughly 660 m, inside the 695 m available. At the deceleration actually achieved it needed 757 m. This shortfall is how the strike reached the overrun. The late reject reached it by a separate route, by shortening the runway available rather than lowering the deceleration. The overrun was one stop with two causes, and neither cause alone carried the aircraft past the end.

It is virtually certain that the deceleration achieved was below the scheduled accelerate-stop performance, and virtually certain that this contributed to the runway overrun.

### **Monitoring of floor proximity escape path lighting**

This matter did not influence the occurrence, but it was found during the investigation and bears on future risk.

While walking back through the cabin during the disembarkation, the first officer noticed sections of the aisle floor strip unlit. A functional check on 24/06/2026, with the emergency lighting armed and the aircraft on battery power, found two of the six floor proximity escape path segments on the left side aft of row 16 dark. The fault was traced to a failed ballast common to both segments.

The defect could not have been found by any check available to the crew or to a line engineer. The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs, but it neither illuminates nor monitors the floor proximity segments. Their only check is a dedicated functional test at the maintenance programme's 24-month interval. The last such test on VH-SBQ had been 11 months before the occurrence, within the interval, so the defect could have remained undetected for up to 13 more months.

The defect had no bearing on this occurrence. No evacuation was commanded, the main cabin lights were on throughout, and the disembarkation by the forward airstair took 4 minutes 8 seconds without urgency. The monitoring gap is nevertheless a standing one. With a fire indication, smoke, or a night evacuation over the wing, two dark segments in the aft cabin could mark the difference between an orderly exit and a blocked one. Between functional tests the state of this escape path lighting on any aircraft of the type is unknown. [NOT ESTABLISHED: whether the failed ballast part is fitted across the fleet, and how many aircraft are currently within the 24-month interval since their last functional test.]

It is virtually certain that the floor proximity escape path lighting on this type is checked only by a functional test at a 24-month interval and not by the pre-flight test, and very likely that this increased risk. It is unlikely that this influenced the occurrence. (Safety Issue)

## **Procedures**

### **No training or guidance for a runway obstruction during the take-off roll**

The objective of module 4 of the recurrent training syllabus TRG-SYL-07 is that the crew "will demonstrate a correctly executed rejected take-off for an engine failure, an engine fire and a configuration warning, each initiated below V1, and will demonstrate the decision to continue for a failure recognised at or after V1." Its four scripted simulator scenarios are an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt, continued. Every one is an aircraft system event. None presents an obstruction, an animal or any other external hazard on the runway, at any speed. In the 2025 to 2026 cycle 214 crew completed the module. No crew has been presented with an external obstruction scenario in the five training cycles on file. Scenario records for earlier cycles were not available, which is why the existence of the gap is not placed higher than extremely likely.

Operations Manual Part B 4.6 has the same gap. Paragraph 4.6.3 lists the items for which a take-off is rejected between 80 kt and V1, and 4.6.4 requires the take-off to be continued at or above V1. The section contains no guidance on an obstruction seen on the runway at any speed. It also contains no amplification of 4.6.4 for a case in which continuing is itself hazardous. Both crew members raised this independently, in almost the same words. Every stop or go scenario they had been given had been an engine, and nobody had ever put something on the runway in front of them in training.

How much this shaped the decision was argued both ways. On one reading, the captain had about a second and nothing behind him. With no scenario and no guidance, the decision on the roll was the first time he had considered the question at all, and preventing that is the purpose of training and procedure. On the other reading, the captain knew the rule and departed from it deliberately. Training on obstructions would not have changed what he knew. He did not reject because he was untrained, but because he judged the alternative worse, and on that view the gap is a finding about the fleet rather than a cause of this reject.

The influence finding is based on the first reading, with the second carried alongside it. An unrehearsed decision made cold in about a second is not the same decision as one made against a briefed and practised frame. A crew who had considered an obstruction above V1 beforehand would have had the question already worked through, whatever answer they had reached. This explains the circumstances in which the decision was made. It does not excuse the decision, and it does not establish what a trained crew would have done.

It is extremely likely that the operator's training and procedures contained no guidance or scenario for a runway obstruction during the take-off roll, and likely that this contributed to the decision to reject the take-off above V1. (Safety Issue)

### **The trigger for a runway inspection after last light**

The operator's Operations Manual Part A 8.14 opens by recognising the hazard: "Wildlife presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk." The control follows at 8.14.2. At an aerodrome listed in Appendix 8A, "the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At other aerodromes, 8.14.3 states that "no specific action is required."

Appendix 8A lists an aerodrome "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nine aerodromes are listed. Tarlton Springs is not among them and never has been. Nothing in Part A admits an aerodrome to the list on any other evidence, and nothing asks a crew to report a sighting. The list was last reviewed on 03/03/2025, four months before the evening service began.

The control is sound in itself, but its trigger is circular. It can only be armed by a reported strike, which is the outcome it exists to prevent. Tarlton Springs had had no company strike, so it could not be listed. Because it was not listed, no inspection was required. Because no inspection was required, none was requested and nothing was looked for. Meanwhile the aerodrome's own log held eleven sightings of kangaroos on the sealed surface. None of them could put the aerodrome on the list, because they were not strikes, were not reported to the company, and no airline had ever asked to see the log.

This is why the omission described under People had no owner. Had 8.14.2 applied at Tarlton Springs, an inspection would have been requested and a pass required before the roll, and aerodrome staff disperse animals when they find them. The only reservation is that an inspection

completed some minutes before departure does not guarantee a clear runway when the aircraft reaches it.

It is virtually certain that the operator required a runway inspection after last light only at aerodromes where a wildlife strike had been reported within 24 months, and very likely that this contributed to no runway inspection being requested before SB512 departed. (Safety Issue)

## Organisation

### The regulatory threshold for a wildlife hazard management plan

Aerodrome Standard AS-139 10.4.1 requires the operator of a certified aerodrome to "take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." A documented wildlife hazard management plan is required under 10.4.2 only where the aerodrome is used by aeroplanes above 27,000 kg maximum take-off weight, or serves more than 350,000 passenger movements in a calendar year. The heaviest scheduled aeroplane at Tarlton Springs is 22,900 kg, and the aerodrome handled 61,400 passenger movements in 2025, so neither limb applies. Under 10.4.3 its measures "need not be documented". This was confirmed against the certified aerodrome register and the aerodrome operator's own compliance file. The aerodrome held no plan, no hazard assessment and no dispersal programme. Beyond the single inspection before the first arrival under Aerodrome Manual 6.3, its only wildlife measure on paper was a sighting log that nobody reviewed. The aerodrome is not in breach of the standard.

The argument for influence runs through the fence. With no plan, nothing at the aerodrome owned the wildlife hazard as a hazard. The failed northern fence sat in the works register as an ordinary maintenance item competing for council money. It was deferred three times, each time as unfunded and for review in the next cycle, with no hazard assessment in front of the decision makers. A documented plan brings a hazard assessment, an owner and an audit against both. Under a plan, the fence would have been a recorded wildlife control whose failure had to be answered for, rather than a deferrable works item.

The argument against is also substantial. Paragraph 10.4.1 still obliged the aerodrome operator to take reasonable measures, and a documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The council's deferrals were budget decisions, and there is no evidence that a documented plan would have been funded any better. A plan does not of itself pay for a fence.

The influence finding is accepted on the balance of the evidence. Its central point is that the deferrals were made with no hazard assessment before the council, and a hazard assessment is exactly what a plan would have supplied. It is held at a lower confidence than most of the other influence findings in this analysis, and the argument against it is not dismissed. This factor bears only on the aerodrome's own controls. The operator's Appendix 8A criterion is the operator's own rule, and no aerodrome plan could have changed it.

It is virtually certain that the threshold in AS-139 10.4.2 left Tarlton Springs with no documented wildlife hazard management plan, and likely that this contributed to the failed northern fence remaining unrepaired. (Safety Issue)

### The failed northern fence

Works register item WR-221, raised in 04/2023, recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers. It noted animals passing through and estimated replacement at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time as unfunded and for review in the next cycle. Item WR-263,

raised in 02/2026, described the same section as deteriorating further. The investigator's site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark (Figure 1). The failure, its known effect and its deferral all appear in the aerodrome operator's own register, and the site visit found the fence in that condition.

The failed section is the physical route by which these animals reached the movement area. The pad at 1,050 m and the carcasses at 1,188 m and 1,221 m are on the same northern side of the runway, and the register has recorded animals passing through this section since the item was first raised. Had it been repaired in 2023, the section from 700 m to 1,300 m would have been intact in 06/2026, and this route would have been closed.

One question cannot be answered on the available evidence. WR-221 recommends replacement but states no specification, and nothing shows that the fence as replaced would have excluded kangaroos, which can clear a stock fence. The pad shows that these two animals came through the gap. It does not show that an intact fence would have stopped them. The sighting log also does not record where any of its 41 kangaroos came onto the aerodrome. For that reason the influence finding is held at likely and no higher.

It is virtually certain that the failed northern fence was left unrepaired through three budget rounds, and likely that this contributed to the two kangaroos reaching the sealed runway.

### **Approval of the evening service without review of the route risk assessment**

Route Risk Assessment RRA-041 issue 3, dated 11/2022, is the current assessment for Tarlton Springs. It assessed the runway length and accelerate-stop margin, a single runway with no lighting redundancy, no alternate within 200 nm at night, limited engineering support, and summer heat and density altitude. It carries the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy." No issue 4 exists.

CMR-2025-118 introduced one evening rotation, Port Carrick to Tarlton Springs to Port Carrick, six days per week from 14/07/2025. It was the first company operation into Tarlton Springs after last light. The change was assessed as "schedule change only", with "No new hazard identified." Against the question of whether a route risk assessment review was required, it recorded "No." It was approved by the Head of Flying Operations on 22/05/2025. The daylight restriction in RRA-041 is not referred to anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department had no involvement beyond receiving the approved change. The approver has since left the company and could not be interviewed, so why the change was assessed in this way is not known.

The restriction, if honoured, would have kept every company departure from Tarlton Springs in daylight. The change process asked whether the change was new rather than whether the existing assessment still held, and it put a company aircraft into the aerodrome after last light in direct contradiction of that restriction. SB512 was scheduled off blocks at 17:20 local and departed on schedule, so it rolled at 17:30, after the end of civil twilight at 17:25, because of the timetable and not because of any delay.

The restriction names night lighting and diversion fuel, not wildlife. A review that reopened RRA-041 might have addressed those two items and approved the evening service subject to lighting and fuel conditions, leaving the departure after last light exactly as it was. Because the approver cannot be asked, what a review would have concluded is inference. The other possibility is more likely. Lifting a daylight-only restriction would have required a review to assess a night operation at an aerodrome with a single runway, no lighting redundancy and no alternate within 200 nm at

night. Approval of the service unchanged by such a review is possible, but it is the less likely of the two outcomes.

It is virtually certain that the evening service was approved without reopening the daylight-only route risk assessment, and likely that this contributed to the take-off roll taking place in darkness. (Safety Issue)

### **Rostering of the evening rotation**

The rostering system shows that the evening rotation operated on 289 days from 14/07/2025 to 18/06/2026. On every one of those days it was rostered as the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00. Those duties were planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a 13 hour 30 minute maximum for five sectors from those sign-on times. On six of those days a member of the rostered crew reported unfit, and the rotation went to a crew already on duty, extending that crew's day from three sectors to five. 18/06/2026 was the sixth such day.

The case that the roster influenced this occurrence is that it places the last sector of the longest duty in the pattern into an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support. When a crew member reports unfit, it gives that sector to a crew whose day has just been extended. On 18/06/2026 it put a crew who described themselves as tired into a night departure 11 hours into their duty.

That case fails on the evidence. The FDR shows the power levers at flight idle 0.6 seconds after the decision and maximum braking at 2 seconds, at the fast end of the fleet's own distribution. There is no error of omission on the roll. The one decision a long duty could plausibly have affected, not requesting a runway inspection, was never put to the crew by any document. The roster cannot be shown to have caused the omission of a step that nobody was asked to take. Whether the crew were fatigued at all is itself unresolved, as discussed under People. Each mechanism through which the roster would have had to act on this occurrence is contradicted by recorded data, and it is unlikely that the roster influenced the occurrence.

The roster remains a standing exposure, present on every day the rotation operates and greatest each time a crew member reports unfit. Every operation of the evening rotation in eleven months has closed a five-sector duty planned to within 45 minutes of the maximum, at an aerodrome with no alternate within 200 nm at night. Six times in the period, a crew's day was extended on the morning of operation from three sectors to five. That exposure is unchanged by the fact that it did not act on this occasion.

It is virtually certain that the evening rotation was rostered to close five-sector duties within 45 minutes of the duty limit, and likely that this increased risk. It is unlikely that this influenced the occurrence. (Safety Issue)

### **Hypotheses tested and excluded**

Two hypotheses were tested and excluded. The first was a tailwind component during the roll and the stop. The aerodrome's automatic observations at 06:30, 07:00 and 08:00 UTC gave a headwind component of 5 to 7 kt on runway 09, the FDR recorded a headwind component of 4 to 6 kt throughout with no tailwind at any point, and the crew's performance calculation used a 5 kt headwind. It is very unlikely that a tailwind existed. The second was a loss of power on the No. 2 engine before the reject decision. Both engines were recorded at 98.4% and 98.5% torque from power set until the power levers were retarded at 07:30:32, with fuel flow, ITT and propeller RPM within normal scatter and no caution, warning or exceedance recorded. The CVR contains no engine-related call before the captain's "Stopping. I have control." The borescope found no pre-

existing damage, and the No. 2 damage and RPM decay both begin at the impact at 07:30:33. It is very unlikely that such a loss existed. The No. 2 engine did lose power, but only as a consequence of the strike, one second after the reject began.

## Findings summary

A finding stated without a likelihood term was established as fact.

### Contributing factors

- It was virtually certain that the threshold in Aerodrome Standard AS-139 section 10.4 left the aerodrome with no documented wildlife plan, and likely that this contributed to the overrun. (Safety Issue)
- The failed northern fence, left unrepaired through three budget rounds, existed with virtual certainty and likely contributed to the overrun. (Safety Issue)
- The departure was very likely within the logged kangaroo peak, from 1 h before to 2 h after last light, and this likely contributed to the overrun.
- With virtual certainty, the operator's procedures required an inspection after last light only where a wildlife strike had been reported in 24 months, and this very likely contributed to the overrun. (Safety Issue)
- It was virtually certain that no runway inspection was requested before departing after last light, and likely that this contributed to the overrun, although no document prompted either crew member to consider one.
- That two kangaroos crossed the strip onto the sealed runway was virtually certain, and it very likely contributed to the overrun.
- It was very likely that no training or guidance existed for a runway obstruction during the take-off roll, and likely that this contributed to the overrun. (Safety Issue)
- The evening service was approved without reopening the daylight-only risk assessment, which was virtually certain, and this likely contributed to the overrun. (Safety Issue)
- The roll in darkness, 5 minutes after last light, with no moon, existed with virtual certainty and likely contributed to the overrun.
- With virtual certainty, the take-off was rejected 6 kt above V1, and this very likely contributed to the overrun.
- The No. 2 propeller and right main gear struck two kangaroos, which removed reverse on No. 2 and contributed to the overrun.
- Deceleration was below the scheduled accelerate-stop performance, approximately 15% below the certified value with reverse on No. 1 only, and this contributed to the overrun.

### Other factors that increased risk

- It was virtually certain that the evening rotation was rostered to close five-sector duties within 45 minutes of the limit, and likely that this increased risk, although it did not contribute to this occurrence. (Safety Issue)
- Floor escape path lighting on the type was checked only every 24 months, not pre-flight, which was virtually certain, and this very likely increased risk, although it did not contribute to the overrun or its outcome. (Safety Issue)

### Other findings

- A tailwind component during the take-off roll and stop was examined and ruled out, because the aerodrome's automatic observations and the recorded flight data both showed a headwind component throughout.

- No. 2 engine power loss before the reject decision was examined and ruled out, because both engines were within 0.1% torque of each other until the power levers were retarded and the No. 2 loss began at the impact, one second after the reject.
- Maximum braking was established within 2 seconds of the reject decision, against a fleet mean of 3.2 seconds, and this was examined and found to reduce risk.

## Appendices

### Appendix A: Test summary for analysed factors

The table below sets out each factor the investigation analysed. For each one it gives the area of the occurrence it concerns, the outcome of each test with the likelihood recorded for it, and the category that resulted.

Each test asked a different question:

- whether the factor existed
- whether it influenced the occurrence
- whether, where it did not influence the occurrence, addressing it would still reduce future risk

A test shown as "Not tested" was not required once an earlier test had settled the category. A test shown as "Inconclusive" was examined and could not be settled, and the factor is not a finding.

| Factor                                                                                                      | Area                      | Existence               | Influence              | Importance | Category                           |
|-------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------|------------------------|------------|------------------------------------|
| The threshold in Aerodrome Standard AS-139 section 10.4 left the aerodrome with no documented wildlife plan | External Influences       | Pass, virtually certain | Pass, likely           | Not tested | Contributing factor (Safety Issue) |
| Failed northern fence left unrepaired through three budget rounds                                           | Organisational Influences | Pass, virtually certain | Pass, likely           | Not tested | Contributing factor (Safety Issue) |
| Evening service approved without reopening the daylight-only risk assessment                                | Organisational Influences | Pass, virtually certain | Pass, likely           | Not tested | Contributing factor (Safety Issue) |
| Inspection after last light required only where a wildlife strike was reported in 24 months                 | Risk Controls             | Pass, virtually certain | Pass, very likely      | Not tested | Contributing factor (Safety Issue) |
| No training or guidance for a runway obstruction during the take-off roll                                   | Risk Controls             | Pass, extremely likely  | Pass, likely           | Not tested | Contributing factor (Safety Issue) |
| Roll in darkness, 5 minutes after last light, with no moon                                                  | Local Conditions          | Pass, virtually certain | Pass, likely           | Not tested | Contributing factor                |
| Departure in the logged kangaroo peak, 1 h before to 2 h after last light                                   | Local Conditions          | Pass, extremely likely  | Pass, likely           | Not tested | Contributing factor                |
| No runway inspection requested before departing after last light                                            | Individual Actions        | Pass, virtually certain | Pass, likely           | Not tested | Contributing factor                |
| Take-off rejected 6 kt above V1                                                                             | Individual Actions        | Pass, virtually certain | Pass, extremely likely | Not tested | Contributing factor                |

|                                                                                      |                           |                                      |                                         |                   |                                                 |
|--------------------------------------------------------------------------------------|---------------------------|--------------------------------------|-----------------------------------------|-------------------|-------------------------------------------------|
| Two kangaroos crossed the strip onto the sealed runway                               | Occurrence Events         | Pass, virtually certain              | Pass, extremely likely                  | Not tested        | Contributing factor                             |
| No. 2 propeller and right main gear struck two kangaroos                             | Occurrence Events         | Pass, virtually certain              | Pass, virtually certain                 | Not tested        | Contributing factor                             |
| Deceleration below the scheduled accelerate-stop performance                         | Occurrence Events         | Pass, virtually certain              | Pass, virtually certain                 | Not tested        | Contributing factor                             |
| Evening rotation rostered to close five-sector duties within 45 minutes of the limit | Organisational Influences | Pass, virtually certain              | Fail, unlikely                          | Pass, likely      | Other factor that increased risk (Safety Issue) |
| Floor escape path lighting checked only every 24 months, not pre-flight              | Risk Controls             | Pass, virtually certain              | Fail, unlikely                          | Pass, very likely | Other factor that increased risk (Safety Issue) |
| Maximum braking established within 2 seconds of the reject decision                  | Individual Actions        | Pass, virtually certain              | Pass, very likely (a reduction in risk) | Not tested        | Positive factor that reduced risk               |
| Flight crew fatigued at the time of the take-off                                     | Local Conditions          | Inconclusive, about as likely as not | Not tested                              | Not tested        | Unresolved, not a finding                       |
| Four northern edge lights above nominal intensity on the night                       | Local Conditions          | Pass, likely                         | Inconclusive, about as likely as not    | Not tested        | Unresolved, not a finding                       |
| Tailwind component during the take-off roll and stop                                 | Local Conditions          | Fail, very unlikely                  | Not tested                              | Not tested        | Excluded                                        |
| No. 2 engine power loss before the reject decision                                   | Occurrence Events         | Fail, very unlikely                  | Not tested                              | Not tested        | Excluded                                        |

## Appendix B: Excluded hypotheses

### Tailwind component during the take-off roll and stop

This hypothesis was examined because the aircraft stopped 62 m past the end of the runway, which raised a question about the wind. It was ruled out. The aerodrome's automatic weather station recorded the wind as 090 degrees at 7 kt at 06:30, 6 kt at 07:00 and 5 kt at 08:00. That is a headwind component of 5 to 7 kt on runway 09. The wind recorded by the FDR during the roll gave a headwind component of 4 to 6 kt throughout, with no tailwind component at any point. The crew's performance entry had used a 5 kt headwind, which matched both sources. No source showed a tailwind at any stage of the roll or the stop.

### No. 2 engine power loss before the reject decision

This hypothesis was examined because a rejected take-off at 118 kt is most commonly an engine event. It was ruled out. The FDR recorded the two engines at 98.4% and 98.5% torque, within 0.1% of each other, from power set at 07:30:02 until the power levers were retarded at 07:30:32. Fuel flow, ITT and propeller RPM on both engines stayed inside normal scatter throughout. No caution, warning or exceedance was recorded at any point before the reject.

The CVR contained no engine-related call. The captain's first words were "Stopping, I have control", one second after the first officer's sighting call. The borescope inspection of both engines found no pre-existing damage. The No. 2 engine did lose power, but that loss began at 07:30:33, at the moment of impact and after the reject had begun. It was a consequence of the strike, not the cause of the reject.

## Appendix C: Sources of evidence

### Interviews and accounts

- The pilot in command and the first officer were each interviewed. Both interviews were recorded and had been completed by 26/06/2026.
- The pilot in command also gave an account in his air safety report.
- During the site visit on 23/06/2026, the aerodrome reporting officer confirmed that no operator had ever asked to see the wildlife sighting log. The reporting officer also confirmed that a runway inspection would have been carried out on 18/06/2026 if one had been requested.

### Recorded data and audio

- Flight Data Section report FDA-2026-0417-R1, issued 24/06/2026, containing the FDR and QAR readout for VH-SBQ, sector SB512.
- CVR-2026-0417 and the Tarlton Springs CTAF recording, covering 07:21 to 07:34 UTC. These were transcribed on 25/06/2026, and a second investigator verified the transcription.

### Weather and environment

- MET-ARCH-SB512-180626, retrieved 19/06/2026, containing the weather observations, the forecast, and the light and moon data.

### Aircraft examination

- Engineering inspection report, VH-SBQ, 19/06/2026 to 21/06/2026, covering the post-occurrence aircraft inspection and runway survey.
- Functional check record, VH-SBQ cabin emergency lighting, 24/06/2026.
- Borescope inspection of both engines.

### Aerodrome

- Site inspection of the northern boundary on 23/06/2026, including photograph SV-2026-0417-07 of the northern perimeter fence at the 1,050 m runway marker (attachments/Failed\_northern\_fence\_left\_unrepaired\_through\_three\_budget\_rounds\_1.jpg).
- Photometric check of the runway 09/27 edge lighting, 25/06/2026.
- Tarlton Springs Aerodrome wildlife sighting log, 01/2025 to 06/2026, copied 23/06/2026.
- Tarlton Shire Council aerodrome works register, items WR-221 (04/2023) and WR-263 (02/2026).
- The aerodrome's daily inspection log for 18/06/2026, and Aerodrome Manual 6.3.
- The aerodrome's electrical log.
- The certified aerodrome register and the aerodrome operator's compliance file.

### Regulatory and operator documents

- Aerodrome Standard AS-139 section 10.4, wildlife hazard management.
- Saltbush Air Operations Manual Part A section 8.14, revision 9, 04/2024, including Appendix 8A.
- Saltbush Air Operations Manual Part B section 4.6, revision 14, 09/2025.
- Recurrent training syllabus TRG-SYL-07, cycle 2025 to 2026, module 4. The scenario records cover the five training cycles on file.
- Route Risk Assessment RRA-041 issue 3 (11/2022), Change Management Record CMR-2025-118 (approved 22/05/2025), and the operator's document register.
- Records of the operator's Safety Department relating to CMR-2025-118.

### Crew records

- Flight crew records and the duty record for 18/06/2026.
- The rostering system extract for the evening rotation, 14/07/2025 to 18/06/2026.
- Drug and alcohol test results for both flight crew.

## Appendix D: Investigation activity

Times in this appendix are UTC.

- 18/06/2026, 09:05: The Duty Operations Controller notified the investigation. The occurrence was assessed as a serious incident that warranted investigation, and the investigation was opened with its scope set out.
- 19/06/2026, 01:20: The recorders were removed at Tarlton Springs and sent to the flight data section. Both flight crew were stood down and tested, with negative results. The Head of Flying Operations suspended the evening Tarlton Springs service.
- 23/06/2026, 08:40: The site visit to Tarlton Springs took place. The northern fence was walked and photographed, and the investigator found a continuous 140 m failure with an animal pad at the 1,050 m mark. The wildlife sighting log was copied and a photometric check of the runway 09/27 edge lighting was requested. The reporting officer confirmed that no operator had ever asked for the sighting log.
- 26/06/2026, 04:15: The interviews with the pilot in command and the first officer were completed. The cabin emergency lighting functional check was raised following the first officer's account.
- 02/07/2026, 06:30: The investigation confirmed that the Appendix 8A listing criterion was self-referential. Only a reported strike at an aerodrome could place it on the list, and no provision asked a crew to report a sighting. The matter was raised with the operator's Safety Department.

## Distribution

| Title                                        | Name        |
|----------------------------------------------|-------------|
| Head of Flying Operations                    | R. Calvert  |
| Head of Safety                               | M. Falkiner |
| Fleet Manager, Turboprop                     | D. Osei     |
| Training Manager                             | S. Arnott   |
| Aerodrome Reporting Officer, Tarlton Springs | K. Pryor    |